

USD/CRC · MONEX — the calendar strategy

2,663 sessions · 2014-12-08 → 2026-06-19 · long/short USD · \$1.00M per trade · priced VWAP-to-VWAP, net of 0.65 CRC round-trip · no technical indicators

One tradeable edge: **Costa Rica's mid-month payment calendar**. Firms sell USD to raise colones for the 15th IVA (D-104) tax filing and the quincena payroll, so the colón strengthens *into* the deadline and reverts after. The rule is **short USD within 6 business days of the deadline, long the rest of the month**, at **half size whenever a slow 20-day volume regime disagrees**. Selection is disciplined — variants are ranked **in-sample** (first 60% of history) and every headline here is the **out-of-sample** realisation on the 40% held out after **2021-11-18**. On that held-out window the rule earns **\$123k/yr at Sharpe 3.55** (full-sample Sharpe 3.31, ~18 trades/yr), and it **strengthens out of sample** (3.22 → 3.55) where the flat-sized entries decay. It is the only edge in the study that works in **both** the pegged 2014–2021 regime and the 2021–2026 float.

3.55

out-of-sample Sharpe

recommended rule · \$123k/yr ·
40% held out

3.22 → 3.55

in-sample → out-of-sample

the edge strengthens on held-out
data, it does not decay

3.31

Sharpe (full sample)

~18 trades/yr · \$1.00M/trade ·
VWAP-priced

3.7 / 3.92

with the exit overlay (IS/OOS)

top of the ranking · max DD \$-16k
(\$2)

52%

BCCR + public sector of
MONEX volume

the official hand marks the mid-
month turn (\$1)

29 vs 18 M

volume: colón up vs down

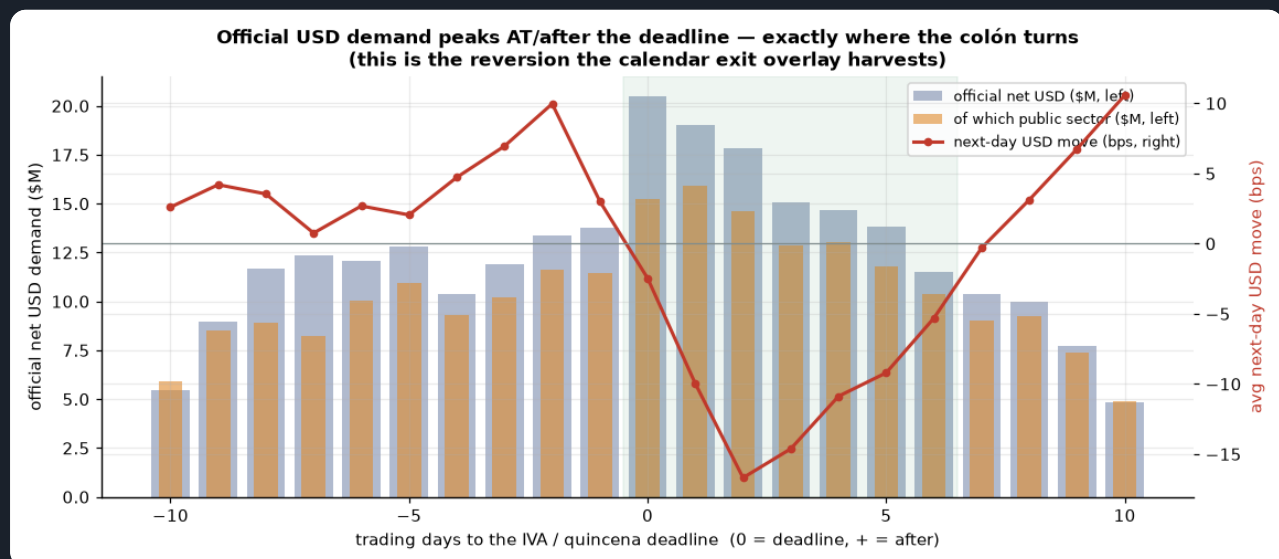
exporters are the swing USD
supply

One accounting basis throughout. Every figure — every table and chart — is priced **VWAP-to-VWAP** next session, **\$1.00M** per trade, net of the **0.65 CRC** round-trip slippage (0.325/side), annualised on the **231** sessions/yr MONEX actually trades — not the 252 equity-market convention. All modules read this from one file (`src/basis.py`), so no two tables can silently disagree; `tests/` reconciles this report against the `out/*.json` source of truth.

1. The mechanism, in one picture

Re-index the price series from raw day-of-month to **business-days-to-the-deadline** and the mechanism is unmistakable. Private firms convert USD → CRC for the 15th, a recurring supply surge that strengthens the colón into the deadline (−9 to −16 bps next-day) and reverts after. The BCCR is not a bystander: official flow — the central bank plus the non-bank public sector (RECOPE et al.) routed through MONEX — is a **median 52%** of daily volume, and official USD demand **peaks exactly at the turn**. Heavy official demand does not push USD up; it marks the point where private mid-month supply overwhelms (contemporaneous corr -0.25). So the calendar reversion and the official footprint are two views of one cash-flow cycle.

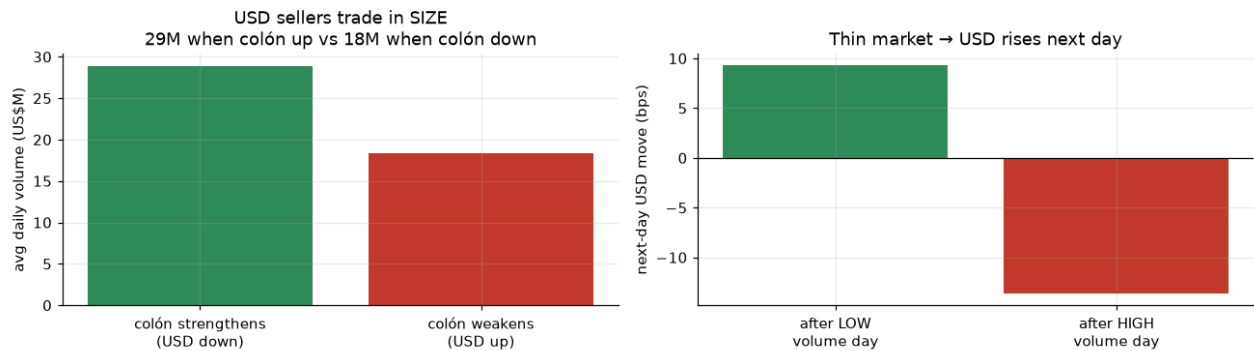
The turn: next-day USD move and official USD demand, by trading-days-to the deadline



Bars: BCCR + public-sector net USD demand (\$M). Line: average next-day USD move (bps). Both are indexed to the statutory IVA (15th) / mid-month payroll deadline, snapped to MONEX's own trading calendar so weekend/holiday rolling is exact. Official demand crests in the shaded window right as the colón strengthens hardest — the reversion the exit overlay (\$2) harvests. The signal decays to -0.08 when lagged one session and depends on same-day disclosure, so it is the mechanism, not a live entry.

Why volume is the size filter: exporters are the swing USD supply

The engine: exporters/inflows are the swing USD supply



On days the colón strengthens the market trades 29M vs 18M when it weakens — USD sellers (exporters, remittances, FDI) come in size, buyers trickle. So a thin market drifts USD up next day (+9.3 bps after quiet days vs -13.7 after busy ones). That asymmetry is why a **slow** volume regime trims size on trades the supply picture does not confirm — without ever flipping the direction.

§2 · THE RULE & WHAT EACH LAYER EARNS

2. Built one design choice at a time

Exact logic

≤ 6 business days before the IVA (D-104) / quincena deadline (through it) →
SHORT USD \$1M
(firms sell USD to raise colones for the 15th tax filing + payroll → colón strengthens)
otherwise → LONG USD \$1M (supply fades, USD drifts back up)
size trim: hold HALF size whenever a slow 20-day volume regime disagrees with the trade
optional exit: bank the trade once it gives back 30 bps from its best (hard floor 40 bps)

Each row adds ONE choice on top of the row above. **IS Sharpe** is the selection metric (first 60%); **OOS Sharpe** is the honest confirmation on the 40% held out after 2021-11-18 and never used to choose. The recommended always-in-market rule is ② **Calendar + slow-vol**; the exit and reserve layers (③–④) are optional **risk** improvements, not more alpha.

BENCHMARK

Always long USD (buy-and-hold)
the naive alternative — dies out of sample

0.83

-1.44

\$26k

\$-111k

THE RULE, ONE DESIGN CHOICE AT A TIME

① Calendar entry — short USD into the deadline
≤6 business days before the IVA/quincena deadline, flat size

3.14

2.79

\$97k

\$-45k

② + slow-volume size trim **RECOMMENDED**
half size when the slow 20-day volume regime disagrees — cuts the fat losers

3.22

3.53

\$76k

\$-24k

③ + trailing 30 / hard floor 40 exit
optional timing overlay — best risk-adjusted, roughly halves drawdown

3.7

3.92

\$101k

\$-16k

④ + BCCR reserve-regime long-trim
trim long-USD size when reserves rise — strictly-causal monthly figure

3.55

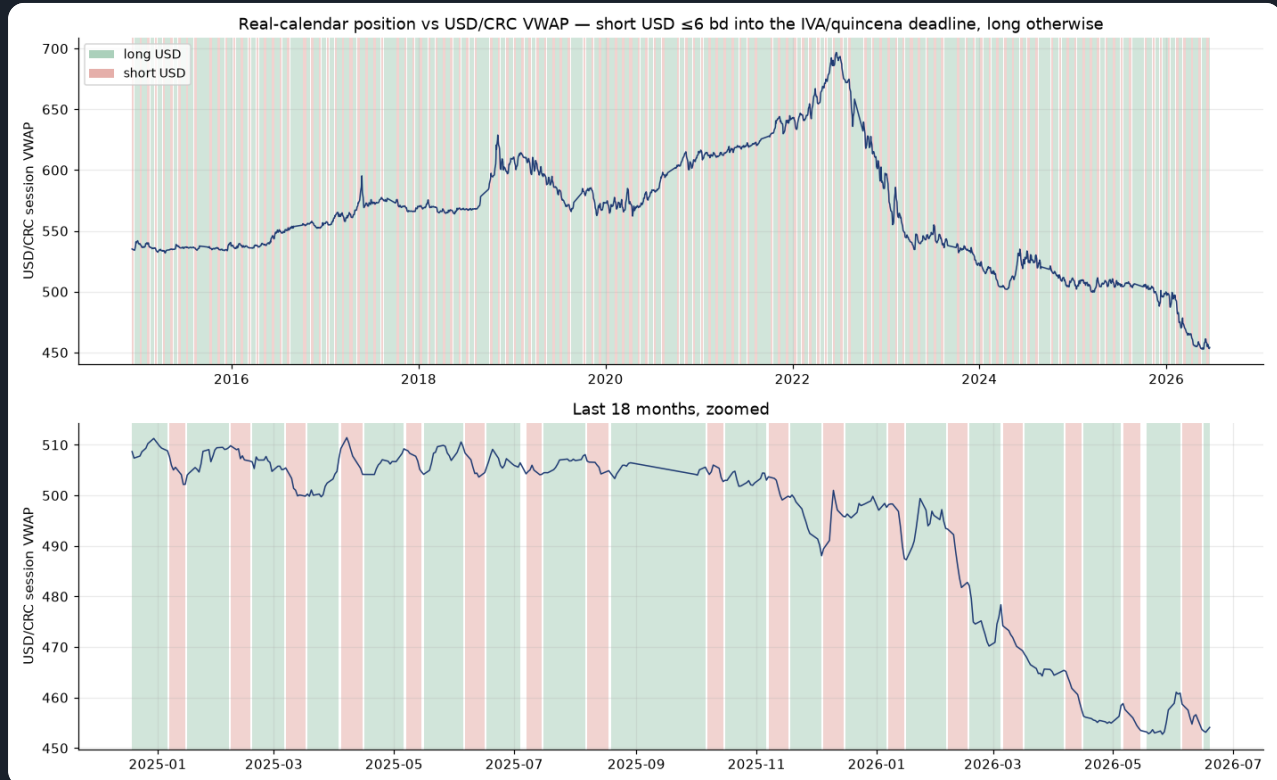
4.09

—

\$-22k

Read the layers as risk, not money. The **slow-vol trim** (②) is what makes it trade-ready — it holds or improves out of sample while flat-sized entries decay, turning out-of-sample per-trade skew positive (**+0.74**) and halving the worst trade (\$-50k → \$-18k). The **exit overlay** (③) is a second, independent tail control: a paired t-test on daily OOS P&L gives $p = 0.7292$ — total P&L is statistically unchanged, but max drawdown drops ~45% and OOS Sharpe rises +1.13. A take-profit actively *hurts* (the edge is a drift into the deadline) and is excluded. The **reserve trim** (④) beats a blind "halve every long" control by \$56k out of sample (paired $t = 3.31$, $p = 0.001$) — the reserve regime picks *which* longs to cut.

What the rule does: session VWAP shaded by the recommended position



Green = long USD, red = short USD (short ≤ 6 business days into the deadline). The rule is always in a position — no flat state — and holds these at half size when the slow volume regime disagrees (same colour, thinner book). The short bands cluster mid-month, exactly where the cash-flow supply surge lands.

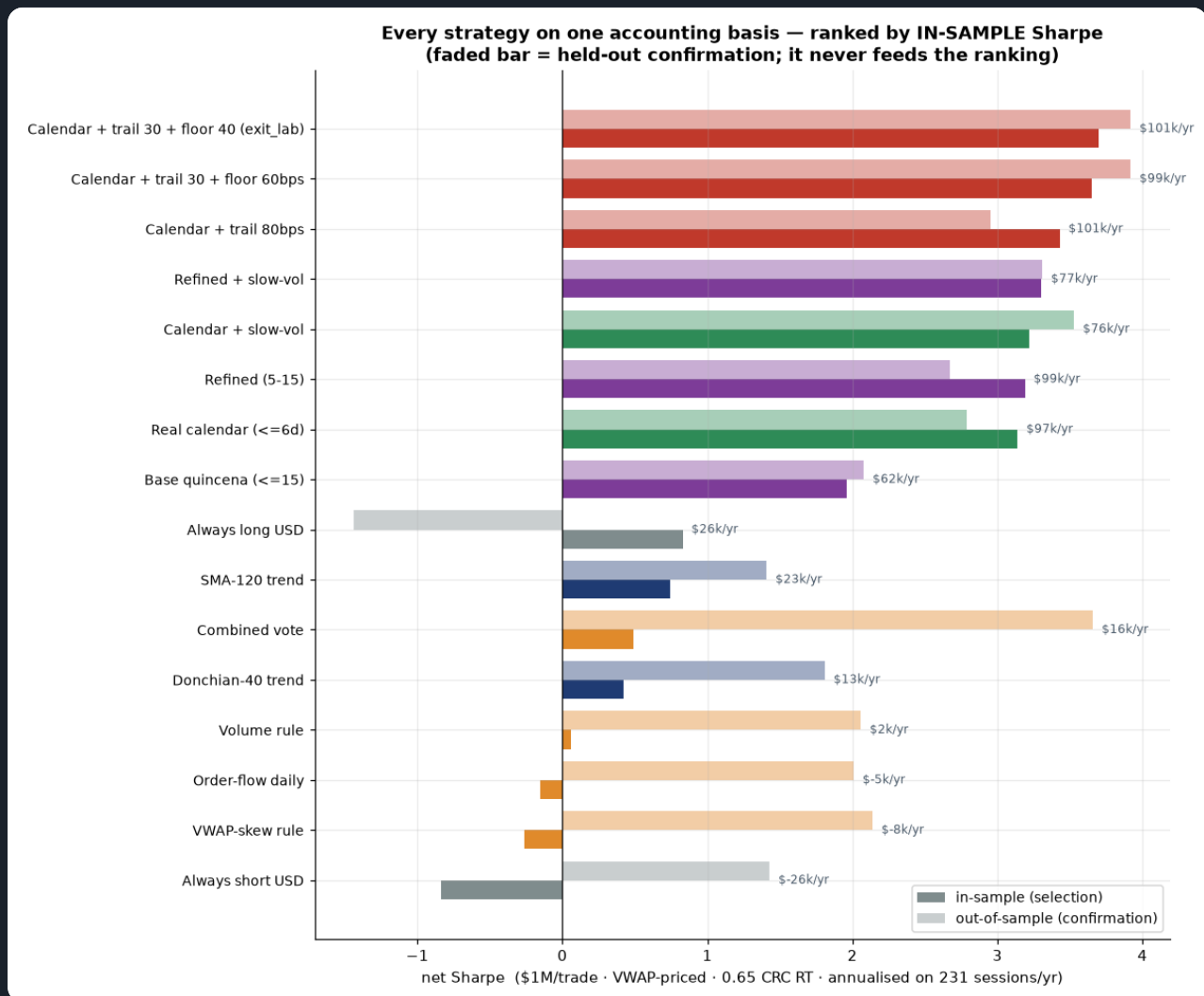
§3 · EVERY STRATEGY ON ONE HONEST BASIS

3. One code path, one ranking — the master comparison

Every rule in the study — the calendar family, the daily flow signals (order-flow, volume, skew), trend books, the ML model, and the two always-on benchmarks — is re-priced through a **single** code path (VWAP-to-VWAP, \$1.00M/trade, 231 sessions/yr) and ranked by **in-sample** Sharpe. This one table and chart replace every per-strategy tearsheet: the calendar family owns the top, the flow family owns the bottom, and the out-of-sample column (carried for honesty, never used to order) tells you which survive.

#	STRATEGY · VWAP-PRICED, NET 0.65 CRC RT, 231 SESS/YR	FAMILY	IS SHARPE	OOS SHARPE	IS \$/YR	IS MAX DD
1	Calendar + trail 30 + floor 40 (exit_lab) TOP OF BOOK	exit	3.7	3.92	\$101k	\$-16k
2	Calendar + trail 30 + floor 60bps	exit	3.65	3.92	\$99k	\$-18k
3	Calendar + trail 80bps	exit	3.43	2.95	\$101k	\$-23k
4	Refined + slow-vol	calendar	3.3	3.31	\$77k	\$-25k
5	Calendar + slow-vol	calendar	3.22	3.53	\$76k	\$-24k
6	Refined (5-15)	calendar	3.19	2.67	\$99k	\$-46k
7	Real calendar (<=6d)	calendar	3.14	2.79	\$97k	\$-45k
8	Base quincena (<=15)	calendar	1.96	2.08	\$62k	\$-55k
9	Always long USD	benchmark	0.83	-1.44	\$26k	\$-111k
10	SMA-120 trend	trend	0.74	1.41	\$23k	\$-66k
11	Combined vote	flow	0.49	3.66	\$16k	\$-278k
12	Donchian-40 trend	trend	0.42	1.81	\$13k	\$-128k
13	Volume rule	flow	0.06	2.06	\$2k	\$-242k
14	Order-flow daily	flow	-0.15	2.01	\$-5k	\$-273k
15	VWAP-skew rule	flow	-0.26	2.14	\$-8k	\$-397k
16	Always short USD	benchmark	-0.84	1.43	\$-26k	\$-195k

Every strategy on one accounting basis, ranked by in-sample net Sharpe



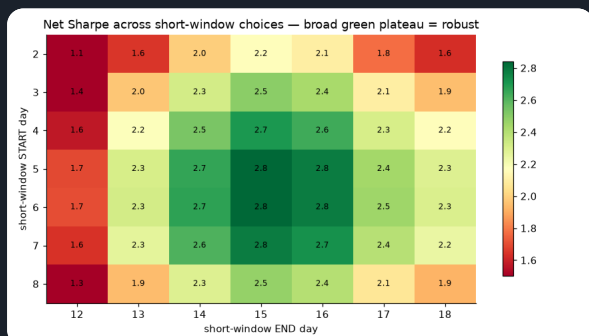
Faded bar = held-out confirmation. The calendar family (green) plus its exit overlays (red) take the top seven seats; the flow rules (orange) sit at the bottom with \$-397k-class drawdowns despite occasionally strong OOS Sharpe — the path is not fundable. The mirror-image benchmarks (Always long IS 0.83 / OOS -1.44, Always short -0.84 / 1.43) confirm the P&L and cost accounting.

The flow signals are regime-contingent. Order-flow, volume, and skew are worthless-to-negative on the pegged 2014–2021 window and only fire on the 2021–2026 float, so their full-sample numbers average a dead regime with a live one — and they carry \$-397k-class in-sample drawdowns at 80–97 round trips/yr. Their value here is only as the *slow filter* that sizes the calendar trade.

Honest caveat. Because in-sample is the pegged/low-vol regime and out-of-sample is the float, in-sample selection structurally favours whatever works in both. The calendar rule wins partly *because* it is that edge — a real virtue, but not the same claim as "the flow signals never work."

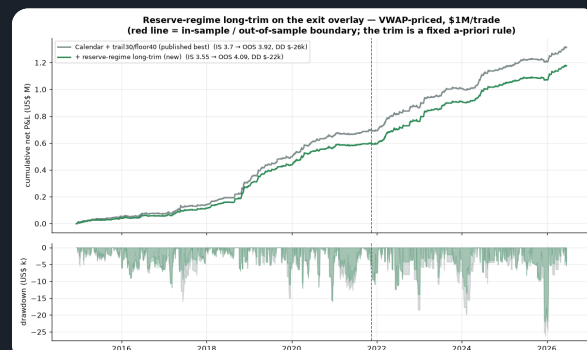
4. A plateau, a clean held-out split, and passed controls

Net Sharpe across every choice of the short-window start & end



A broad plateau (Sharpe 1.05–2.84), not a lucky point — days 5 → 15 is simply the natural mid-month supply window and every neighbour works. The ≤ 6 -day anchor, 20-day volume window and half-size trim are fixed a-priori, not fitted.

Held-out equity: the exit overlay + reserve trim vs the plain rule



Red line = the in-sample / out-of-sample boundary; nothing right of it is fitted. Both curves compound through the boundary and the overlaid rule's drawdowns (lower panel) stay visibly shallower, most clearly through the deep late-sample selloff.

Held out, the edge grows. The recommended rule goes IS 3.22 → OOS 3.55; the exit overlay IS 3.7 → OOS 3.92 — no decay. Across the exit parameter grid $\text{corr}(\text{IS}, \text{OOS}) = +0.82$, so in-sample ranking genuinely predicts out-of-sample, and the deflated Sharpe (Bailey / López de Prado) over 391 trials gives $P(\text{true} > 0) \approx 1.0$.

Positive every year. The rule is positive in all 12 calendar years (worst \$34k), including the 2015–17 pegged years where the volume signal was dead. Priced at the closing print instead of the VWAP the edge slightly *improves* — it is an economic flow, not an artifact of marking to the close.

§5 · VERDICT & RISKS

Trade the calendar + slow-vol rule. Short USD into Costa Rica's mid-month IVA / payroll deadline (within 6 business days), long the rest of the month, at half size when a slow 20-day volume regime disagrees. Full-sample \$95k/yr at Sharpe 3.31, ~18 trades/yr — and, the number that matters, **out-of-sample Sharpe 3.55** (\$123k/yr on the 40% held out).

The optional overlays are risk, not alpha. A trailing 30 / hard-floor 40 exit tops the ranking (IS 3.7 → OOS 3.92) and roughly halves drawdown, and a BCCR reserve-regime long-trim lifts

OOS Sharpe to 4.09 — both smoother paths to size up, with total P&L statistically unchanged. Add them if you want the drawdown, skip them for simplicity.

The mechanism is economic, not technical: exporter USD supply and the tax/payroll cycle. That is why it persists across the peg and the float — and why the **dominant, unpriceable risk is a re-peg or heavy BCCR intervention**: the bank is already ~half of MONEX and could mute the edge at will. The reserve trim is a partial hedge, not an escape.

Limits. The float that carries most of the edge is only the last ~4.5 years; results are net of VWAP slippage but gross of financing/borrow and market impact; the colón is not freely shortable for all participants — this is a directional/treasury edge, not a market-neutral one.

Dollar & pricing convention. \$1.00M USD notional per full position; 1 bp of next-session move = \$100; slippage 0.325 CRC/side charged on every position change. **All P&L is priced VWAP-to-VWAP** — a position decided from information through session t earns the next session's VWAP move — net, gross of financing/borrow and market impact, non-compounded (reset to \$1.00M each day). **Selection discipline:** variants are ranked on the in-sample Sharpe (first 60%, 2014-12-08 → 2021-11-17); every headline result is the out-of-sample realisation on the held-out 40% (2021-11-18 → 2026-06-19). Nothing is re-fit on the test window. Every part is annualised on the venue's real 231 sessions/yr from a single basis (`src/basis.py`). Inputs: one BCCR MONEX price/volume export plus two BCCR official-flow exports (FX intervention `CodCuadro 1587`, net reserves `CodCuadro 8`). Reproducible: `parse_monex` → `parse_bccr` → `analyze` → ... → `rank_strategies` → `exit_lab` → `intervention` → `build_report`. See **FINDINGS.md** for the full written record.